Kernel Corridors and Schreier Waiting in Synchronizing Automata

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This paper is Paper XXIII of the RIME program. It develops exact rank-escape, marked-kernel corridor, and Schreier-waiting theorems for declared classes of synchronizing automata.

Abstract

Problem. A synchronizing automaton reduces the rank of its current image subset only when a word merges at least one surviving pair. Classical compression bounds control how long this can take without retaining the marked transformation kernel, the collision capacity of the exit, or the permutation geometry of a rank-preserving corridor.
Approach. We separate rank escape into four exact objects: pair-hitting distance, reachable packing, marked-kernel corridors, and fiber incidence. On the unit-defect branch, an inverse-closed permutation subalphabet acts inside one marked-kernel stratum. A local letterwise simulation condition normalizes every shortest plateau escape to a path in a subset Schreier graph followed by one defect-one cover exit.
Results. For a reachable subset T, its rank-escape time is the minimum pair-hitting distance inside T. A rank-preserving plateau is equivalently a suffix ball confined to one marked-kernel stratum; on the unit branch its first exit is a partition-lattice cover. Reachable-unit packing is both the exact discrete inverse and the exact layer-cake representation of the unit waiting tail. An exact rankwise accounting interface gives

$$D_{\mathrm{sync}} \leq \Phi_{\mathrm{FI}}(Q) \leq \sum_{j=2}^{n} \max\{\bar{u}_j, \bar{h}_j\},$$

separating unit waiting from high-capacity escape. For an inverse-closed permutation core G, unit waiting is bounded by a Schreier cover radius. If $G \cong (\mathbf{F}_2)^s$ is generated by commuting involutions, then

$$D_{\mathrm{sync}} \leq 1 + (n-2)(s+1).$$

Faithfulness of $G \leq S_n$ gives $s \leq \lfloor n/2 \rfloor$, hence the parameter-free bound

$$D_{\mathrm{sync}} \leq 1 + (n-2)\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right) \leq (n-1)^2.$$

Thus the declared commuting-involutive class satisfies the Černý threshold. In the binary case, every shortest normalized reset word has

$$D_{\mathrm{sync}} = n - 1 + \nu(A),$$

where $\nu(A)$ counts genuine permutation waits. Under clean exit, a hole-set recurrence constructs a reset word with at most one wait. Consequently

$$D_{\mathrm{sync}} \in \{n-1, n\},$$

and the exact clean-class extremal depth is n for every $n \geq 3$.

Boundary. These are class theorems under explicit defect-one, inverse-closure, simulation, commutativity, and clean-exit hypotheses. They do not prove the Černý conjecture for arbitrary synchronizing automata. Exact finite censuses replay the declared certificates but are not used as theorem proofs.

Keywords: synchronizing automata; Černý conjecture; transformation kernels; pair automata; Schreier graphs; elementary abelian permutation groups; defect-one maps; reset threshold.

Notation Table

Symbol	Meaning
$A = (Q, \Sigma, \delta)$	complete deterministic finite automaton
$n = \|Q\|$	number of states
$T.w$	image of $T \subseteq Q$ under the word w
t_w	transformation of Q induced by w
$\mathcal{R}_r$	reachable subsets of cardinality r
D_{sync}	shortest reset-word length
$\omega(T)$	shortest strict rank-escape length from T
$d_2(x, y)$	shortest word length merging the pair $\{x, y\}$
$P_A(L)$	maximum pair-distance packing above radius L
$\text{KerPart}(t_w)$	fiber partition of the transformation t_w
$e^*(T)$	best terminal excess among shortest exits from T
θ_r, χ_r, κ_r	exact, statewise, and layerwise incidence controls
u_r, $\bar{u}_j$	unit waiting profile and its rank tail
h_r, $\bar{h}_j$	high-capacity ratio profile and its rank tail
$P^{\text{reach}}_{A,U}(L)$	reachable-unit packing profile
Π, $G = \langle \Pi \rangle$	permutation core alphabet and generated group
$\partial_1 \mathcal{G}^{\Pi}_K$	defect-one cover boundary in a kernel Schreier orbit
s, s_T	global elementary-abelian rank and subset quotient rank
$\nu(A)$	minimum genuine permutation-wait number

1 Introduction

Let $A = (Q, \Sigma, \delta)$ be a synchronizing automaton. A word w resets A when $|Q.w| = 1$. Since word action can only coarsen the transformation kernel, every reset trajectory has a rank filtration

$$n = r_0 \geq r_1 \geq \cdots \geq r_d = 1.$$

Throughout, $n = |Q| \geq 2$.

The number of strict drops is at most $n - 1$. The main difficulty is the waiting between them.

The usual rank-layer question asks how long a reachable r-subset can avoid all collisions. It separates into two exact chains:

$$\boxed{\text{rank escape} \longleftrightarrow \text{pair hitting} \longleftrightarrow \text{packing} \longleftrightarrow \text{marked-kernel corridors}}$$

and, on a structured unit-defect class,

$$\boxed{\text{kernel corridor} \longrightarrow \text{subset Schreier graph} \longrightarrow \text{cover radius} \longrightarrow \text{reset bound.}}$$

The first chain is valid for every finite deterministic automaton. The second requires a declared inverse-closed permutation core and defect-one nonpermutation letters. It provides a group-theoretic mechanism for controlling unit waiting.

The strongest specialization occurs in the binary clean class. There the rank-preserving core consists of one involution p, while the defect-one map d has one collision pair and one missing image state. Passing from a subset T to its hole set $H = Q \setminus T$ turns rank descent into a growing-set recurrence. A second plateau on the canonical descent forces a closed two-subset trap. This gives an exact all-n extremal result rather than a fitted sequence.

1.1 Main theorem spine

The main results are:

1. **Pair-hitting identity.** For every nonsingleton subset T,
$$\omega(T) = \min_{\{x,y\} \subseteq T} d_2(x, y).$$
2. **Waiting-capacity interface.** The exact rankwise accounting interface obeys
$$D_{\text{sync}} \leq \Phi_{\text{FI}}(Q) \leq \sum_{j=2}^{n} \max\{\bar{u}_j, \bar{h}_j\}.$$
3. **Marked-kernel corridor theorem.** If $T = Q.w$ and $K = \text{KerPart}(t_w)$, then
$$\omega(T) > L \iff \text{KerPart}(t_{wv}) = K \quad \text{for every } |v| \leq L.$$
4. **Reachable-unit packing identity.** The reachable-unit packing profile is the discrete inverse of the unit waiting tail and satisfies
$$\sum_{j=2}^{n} \bar{u}_j = \sum_{L \geq 0} \left(P_{A,U}^{\text{reach}}(L) - 1 \right)_+ .$$
5. **Schreier normalization theorem.** Under letterwise Schreier simulation,
$$\omega(T) = 1 + \text{dist}_{\mathcal{G}_K^\Pi}(T, \partial_1 \mathcal{G}_K^\Pi).$$
6. **Commuting-involutive core theorem.** If $G = \langle \Pi \rangle \cong (\mathbf{F}_2)^s$, then
$$D_{\text{sync}} \leq 1 + (n-2)(s+1).$$
7. **Parameter-free Černý corollary.** Faithfulness gives $s \leq \lfloor n/2 \rfloor$, and hence the declared class satisfies $D_{\text{sync}} \leq (n-1)^2$.
8. **Binary clean one-wait theorem.** A binary clean synchronizing automaton has $\nu(A) \leq 1$ and $D_{\text{sync}} \in \{n-1, n\}$. Its exact extremal depth is n for every $n \geq 3$.

1.2 Novelty and claim boundary

Pair automata, transformation semigroups, Schreier graphs, elementary abelian permutation groups, and the Černý problem are established subjects [18, 19]. The new interface among them consists of marked-kernel plateau semantics, local simulation into a permutation-core Schreier graph, the statewise quotient-rank bound, and the hole-recurrence obstruction to a second clean binary wait. Section 1.3 compares this interface with the closest defect-one, permutation-group, and pair-digraph results.

The classical Černý problem and the Pin–Frankl compression estimate are background and are not reproved as new results [7, 9, 12]. Zhu's 2026 arXiv preprint on one-cluster automata and its spectral methods [22] is a comparison result and is not used in subsequent proofs. The standard Černý family with more than two states does not satisfy the inverse-closed alphabet condition required by the subclass theorems, so none of those results proves the general conjecture.

1.3 Related work

Defect-one and almost-group automata. Catalano and Jungers study automata formed from permutation letters and one rank-$(n-1)$ merging letter, and construct families with reset threshold $\Omega(n^2/4)$ [6]. Their result shows that the raw permutation-plus-defect-one alphabet format alone cannot imply these bounds. Berlinkov and Nicaud study synchronization and recognition for almost-group automata, whose unique nonpermutation letter acts as a permutation on all but one state [2]. Bondar, Casas, and Volkov develop structural characterizations of completely reachable automata [3], while Casas and Volkov characterize the binary case [4]. Subsequent work gives quadratic reaching and reset bounds on completely reachable classes [8], but these results do not yield the binary clean one-wait theorem of Section 6. Casas Torres treats completely reachable almost-group automata with one defect-one letter and a transitive permutation core [5]. The binary characterization also shows, apart from the two-state flip-flop, that in the binary permutation-plus-defect-one setting complete reachability forces the permutation letter to be an n-cycle. In the binary clean class, inverse closure gives $p^2 = 1$. Hence for $n > 2$ the binary clean class is disjoint from that completely reachable class. The same binary separation holds from Casas Torres's transitivity hypothesis: the group $\langle p \rangle$ has order at most two and cannot act transitively on more than two states. Thus these papers concern not only a different conclusion but, on the nontrivial binary surface, a different class.

Permutation-group reset bounds. Rystsov's work on commutative and solvable automata and on simple idempotents gives important structured Černý classes [14, 15, 16]. Zhu proves the bound $2n^2 - 7n + 7$ when every letter has rank at least $n - 1$ and the bijective letters generate a transitive permutation group [21]. The commuting-involutive theorem does not assume transitivity, but it does assume inverse closure, defect-one nonpermutation letters, letterwise Schreier simulation, and an elementary abelian permutation core. Its sharper numerical conclusion therefore holds on a different, narrower structural surface rather than improving Zhu's theorem on Zhu's full class. For the binary exact theorem and $n > 2$, Zhu's transitivity hypothesis is impossible because the core is generated by one involution. The multi-generator theorem can overlap Zhu's class, since an elementary abelian 2-group can act transitively, even regularly.

Rystsov's more recent Černý-type theorem treats a simple idempotent together with a regular permutation group and proves the Černý and rank conjectures on that class [13]. Results surveyed by Araújo, Cameron, and Steinberg likewise cover regular group-action surfaces, including elementary-abelian Černý-group cases [1]. Consequently, the numerical parameter-free Černý conclusion of Theorem 5.4 is not claimed as new on the regular elementary-abelian overlap. The distinct surface is the statewise marked-kernel/quotient-rank formula, its applicability to faithful nonregular or intransitive actions, and the binary clean exact theorem. In the binary case and $n > 2$, a group generated by one involution cannot act regularly, so the exact $n - 1/n$ theorem lies outside Rystsov's Černý-type class.

One-cluster automata. Zhu's July 2026 arXiv preprint presents a proof of the Černý conjecture for one-cluster automata [22]. If the distinguished one-cluster letter has unique cycle length m and positive level ℓ, the claimed parameter-dependent reset bound is

$$(m - 1)(n - 1) + m\ell \leq (n - 1)^2.$$

The proof uses finite-dimensional linear algebra and annular spectral descent, independently of the marked-kernel and hole-recurrence methods in Sections 3 and 6. Its short relative-extension theorem proves the positive-level part of Conjecture 1 of Kisielewicz, Kowalski, and Szykuła [11]. The two structural classes are not nested. A binary clean defect-one map may have several cycles and hence need not be one-cluster; Zhu's theorem, in the other direction, requires no inverse-closed permutation core, letterwise Schreier simulation, or clean exit. The conclusions are correspondingly different: Zhu's preprint presents the Černý threshold on the one-cluster class, whereas Section 6 obtains the exact $n-1/n$ depth dichotomy on the narrower binary clean class.

Pair digraphs and group diameter. Gonze et al. connect reset thresholds with pair-digraph diameters of permutation automata and with permutation-group diameter problems [10]. That work is the closest geometric background for the pair-hitting and group-diameter viewpoint. The additional object in the marked-kernel argument is a reachable subset together with its marked transformation kernel. The resulting distance is to a defect-one cover boundary inside a subset Schreier orbit, not merely a diameter of the underlying state-pair digraph.

Within the cited comparison set, no theorem was found that specializes under the inverse-closed binary clean hypotheses to the one-wait obstruction or the exact reset-depth dichotomy $D_{\text{sync}} \in \{n-1, n\}$. The comparison includes Volkov's 2026 class list [20] and Szykuła's open-problem survey [17]. This is a bounded literature statement, not an exhaustive novelty claim.

The missing implication in the checked specializations is supplied by the cross-layer no-two-waits theorem: after the first genuine permutation wait, any later nonsingleton plateau forces a closed hole-trap incompatible with synchronization.

2 Deterministic Rank Escape

2.1 Subset action and synchronization

Extend δ to subsets and words by

$$T.a = \{\delta(x, a) : x \in T\}, \qquad T.(wa) = (T.w).a.$$

The reachable subset orbit is

$$\mathcal{O}(Q) = \{Q.w : w \in \Sigma^*\},$$

and its rank-r layer is

$$\mathcal{R}_r = \{T \in \mathcal{O}(Q) : |T| = r\}.$$

For any nonsingleton subset define

$$\omega(T) = \min\{|v| : |T.v| < |T|\},$$

with value ∞ if no such word exists. In a synchronizing automaton every reachable nonsingleton has finite escape.

The rank-preserving graph on $\mathcal{R}_r$ retains the labelled edge $T \xrightarrow{a} T.a$ exactly when $|T.a| = r$. Its collision boundary is

$$\partial\mathcal{R}_r = \{T \in \mathcal{R}_r : |T.a| < r \text{ for some } a \in \Sigma\}.$$

Directly from shortest paths,

$$\omega(T) = 1 + \operatorname{dist}(T, \partial\mathcal{R}_r).$$

2.2 Pair-hitting distance

For distinct $x, y \in Q$, define

$$d_2(x,y) = \min\{|w| : \delta(x,w) = \delta(y,w)\},$$

again allowing ∞.

Theorem 2.1 (Pair-Hitting Identity)**.**
For every subset T with $|T| \geq 2$,

$$\omega(T) = \min_{\{x,y\}\subseteq T} d_2(x,y)$$

in the extended nonnegative integers.

Proof. A word drops the rank of T exactly when its restriction to T is noninjective, equivalently when it merges at least one pair in T. Therefore every rank-dropping word has length at least the displayed minimum. Conversely, a shortest word merging a pair that realizes the minimum is noninjective on T and hence drops its rank.

Thus

$$\Omega_r := \max_{T\in\mathcal{R}_r} \omega(T) = \max_{T\in\mathcal{R}_r} \min_{\{x,y\}\subseteq T} d_2(x,y).$$

Rank escape is therefore an exact pair-distance packing problem, not merely analogous to one.

For $L \geq 0$, define the unrestricted automaton-specific packing number

$$P_A(L) = \max\{|T| : d_2(x,y) > L \text{ for all distinct } x, y \in T\}.$$

For a word w, let t_w be its transformation of Q and let $\mathrm{KerPart}(t_w)$ be the partition into its nonempty fibers; $\mathrm{rank}(t_w) = |Q.w|$.

Proposition 2.2 (Packing and Kernel Common Transversals)**.**
A subset T is feasible in the definition of $P_A(L)$ if and only if it is a partial transversal of $\mathrm{KerPart}(t_w)$ for every word w with $|w| \leq L$. Consequently,

$$P_A(L) \leq \min_{|w|\leq L} \mathrm{rank}(t_w),$$

and

$$r > P_A(L) \Longrightarrow \Omega_r \leq L.$$

Proof. A subset is a partial transversal of $\mathrm{KerPart}(t_w)$ exactly when t_w is injective on that subset, or equivalently when no pair in the subset is merged by w. Requiring this for every word of length at most L is exactly the pair-distance condition in the definition of $P_A(L)$. A partial transversal has at most one point in each kernel block, proving the rank bound. If $r > P_A(L)$, every r-subset has a pair at distance at most L; Theorem 2.1 then gives the last implication.

The unrestricted packing number can lose reachability and marked-prefix information. Section 3 restores both on the unit branch.

2.3 Fiber excess and unit exits

For a letter a, let $\mathcal{K}(a)$ be its fiber partition. Define

$$\operatorname{Exc}_a(T) = \sum_{B \in \mathcal{K}(a)} \max(|T \cap B| - 1, 0).$$

Counting all but one point in each nonempty fiber intersection gives

$$|T| - |T.a| = \operatorname{Exc}_a(T).$$

Every shortest escape has the form $v = ua$, where every prefix through u is rank preserving and the last letter a drops rank. Define

$$e^*(T) = \max\{\operatorname{Exc}_a(T.u) : v = ua,\ |v| = \omega(T),\ |T.v| < |T|\}.$$

The **unit branch** consists of states with $e^*(T) = 1$. These are the extremal states for which a shortest wait buys only one unit of rank collapse.

2.4 The waiting-capacity interface

Fix $T \in \mathcal{R}_r$. For every rank-preserving path from T to $U \in \mathcal{R}_r$ followed by a dropping letter a, charge the ratio

$$\frac{\operatorname{dist}_r(T, U) + 1}{\operatorname{Exc}_a(U)}.$$

Let $\theta(T)$ be the minimum of these ratios, and put

$$\theta_r = \max_{T \in \mathcal{R}_r} \theta(T).$$

All quantities below are restricted to finite-escape active layers; in a synchronizing automaton every reachable nonsingleton layer is active. Define

$$\Omega_r = \max_{T \in \mathcal{R}_r} \omega(T), \qquad \varepsilon_r = \min_{T \in \mathcal{R}_r} e^*(T),$$

and

$$\chi_r = \max_{T \in \mathcal{R}_r} \frac{\omega(T)}{e^*(T)}, \qquad \kappa_r = \frac{\Omega_r}{\varepsilon_r}.$$

A shortest escape from T is one of the competitors defining $\theta(T)$. Choosing among shortest escapes one with terminal excess $e^*(T)$ gives

$$\boxed{\theta_r \le \chi_r \le \kappa_r.}$$

The second inequality is statewise: $\omega(T) \le \Omega_r$ and $e^*(T) \ge \varepsilon_r$.

To keep the two mechanisms separate, define

$$u_r = \max_{\substack{T \in \mathcal{R}_r \\ e^*(T)=1}} \omega(T), \qquad h_r = \max_{\substack{T \in \mathcal{R}_r \\ e^*(T)>1}} \frac{\omega(T)}{e^*(T)},$$

with empty maxima equal to zero. Then

$$\chi_r = \max\{u_r, h_r\}.$$

Let

$$\lambda_j = \max_{r \geq j} \theta_r, \qquad \bar{u}_j = \max_{r \geq j} u_r, \qquad \bar{h}_j = \max_{r \geq j} h_r,$$

and define the rank potential

$$\Phi_{\mathrm{FI}}(T) = \sum_{j=2}^{|T|} \lambda_j.$$

Theorem 2.3 (Waiting-Capacity Interface)**.**
Every synchronizing automaton satisfies

$$\boxed{D_{\mathrm{sync}} \leq \Phi_{\mathrm{FI}}(Q) \leq \sum_{j=2}^{n} \max\{\bar{u}_j, \bar{h}_j\}.}$$

Proof. At a rank-r state, choose a same-rank path and terminal dropping letter attaining $\theta(T)$. If the terminal excess is e, the segment has length $e\theta(T) \leq e\theta_r$. The potential drop across ranks r through $r - e$ is

$$\sum_{j=r-e+1}^{r} \lambda_j \geq e\theta_r,$$

because each tail maximum in that range contains θ_r. Thus the selected escape segment is paid for by the potential decrease. Concatenating until rank one proves the first inequality. The relations

$$\lambda_j \leq \max_{r \geq j} \chi_r = \max\{\bar{u}_j, \bar{h}_j\}$$

prove the second.

Theorem 2.3 is an exact rankwise accounting interface, not a proposed universal Černý potential. In particular, no inequality $\Phi_{\mathrm{FI}}(Q) \leq (n-1)^2$ is claimed. The theorem separates unit waiting from high-capacity escape; controlling the pathwise compatibility of those contributions is a separate problem.

3 Marked-Kernel Corridors and Packing Inversion

3.1 Kernel persistence

Let $t_w : Q \to Q$ be the transformation induced by w, and let

$$\mathrm{KerPart}(t_w) = \{t_w^{-1}(z) : z \in \mathrm{im}(t_w)\}$$

be its partition into nonempty fibers. Appending a word can only coarsen this partition.

Theorem 3.1 (Marked-Kernel Corridor)**.**
Let $T = Q.w$ and $K = \mathrm{KerPart}(t_w)$. For every $L \geq 0$,

$$\omega(T) > L \iff \mathrm{KerPart}(t_{wv}) = K \quad \textit{for every word } v \textit{ with } |v| \leq L.$$

Proof. The blocks of $\mathrm{KerPart}(t_{wv})$ are obtained by coarsening K, and their number equals $|T.v|$. Thus $|T.v| = |T|$ exactly when the coarsening has the same number of blocks as K, which forces equality of the two partitions. The condition $\omega(T) > L$ says precisely that this happens for every suffix of length at most L.

For the fixed automaton A, the numerical escape time $\omega(T)$ is determined by the image subset T. The marked kernel does not add a second waiting coordinate; it records the upstream fiber structure that remains unchanged throughout the corridor and the structure of the eventual kernel coarsening. This is stronger than saying only that a pair has not yet hit the diagonal.

Corollary 3.2 (Unit Kernel Cover)**.**

If $e^*(T) = 1$, then every best shortest exit from the marked-kernel corridor changes K by a cover in the partition lattice: exactly two blocks merge and all other blocks remain unchanged.

Proof. Appending the exit coarsens K. Unit excess reduces the number of blocks by exactly one. A strict coarsening with one fewer block merges exactly two old blocks.

Upstream pair mass. A unit kernel cover is unit only at the image-rank level. If the two merged marked-kernel blocks are B and C, then the cover newly identifies exactly $|B||C|$ cross-block source-state pairs. Thus unit terminal excess does not imply unit upstream pair mass.

3.2 Reachable-unit packing

For $L \geq 0$, define

$$P^{\mathrm{reach}}_{A,U}(L) = \max\{|T| : T \in \mathcal{O}(Q),\ e^*(T) = 1,\ \omega(T) > L\},$$

with maximum zero when the set is empty. Recall u_r and $\bar{u}_j$ from Section 2.4, with empty maxima equal to zero.

Proposition 3.3 (Reachable-Unit Packing Inversion)**.**

For every $j \geq 2$ and $L \geq 0$,

$$P^{\mathrm{reach}}_{A,U}(L) < j \iff \bar{u}_j \leq L.$$

Consequently,

$$\bar{u}_j = \min\{L \geq 0 : P^{\mathrm{reach}}_{A,U}(L) < j\}.$$

Proof. The inequality on the left says that no reachable unit state of rank at least j has waiting time greater than L. This is exactly the right-hand condition. The inverse formula follows because the quantities are integer-valued.

Corollary 3.4 (Reachable-Unit Packing Area)**.**

With $(x)_+ = \max\{x, 0\}$,

$$\boxed{\sum_{j=2}^{n} \bar{u}_j = \sum_{L \geq 0} \left(P^{\mathrm{reach}}_{A,U}(L) - 1\right)_+ .}$$

Proof. Since the $\bar{u}_j$ are nonnegative integers,

$$\sum_{j=2}^{n} \bar{u}_j = \sum_{L \geq 0} \#\{j \in \{2, \dots, n\} : \bar{u}_j > L\}.$$

By Proposition 3.3, $\bar{u}_j > L$ if and only if $P^{\mathrm{reach}}_{A,U}(L) \geq j$. The inner count is therefore $(P^{\mathrm{reach}}_{A,U}(L) - 1)_+$.

The packing profile is therefore both the exact discrete inverse and the exact layer-cake representation of the unit waiting tail.

4 Inverse-Closed Permutation Cores

4.1 Structural class

Partition the alphabet as

$$\Sigma = \Pi \sqcup \mathcal{D},$$

where every letter in Π is a permutation of Q, Π is nonempty and inverse-closed as a labelled alphabet, and every $d \in \mathcal{D}$ has global rank $n - 1$. A defect-one letter has one doubleton fiber, so every strict drop it induces on a subset is a unit drop. Hence in a synchronizing automaton of this class every reachable nonsingleton state lies on the unit branch.

Fix a reachable state $T = Q.w$ and its marked kernel $K = \mathrm{KerPart}(t_w)$. Postcomposition by a permutation preserves K. The Π-orbit of T is therefore the image projection of a Schreier graph inside one marked-kernel stratum. Write this graph as $\mathcal{G}^{\Pi}_K$. Its cover boundary is

$$\partial_1 \mathcal{G}^{\Pi}_K = \{U \in \mathrm{Orb}_{\Pi}(T) : |U.d| < |U| \text{ for some } d \in \mathcal{D}\}.$$

4.2 Three normalization levels

The following conditions must remain distinct.

Clean exit. For every reachable nonsingleton state U and defect letter d,

$$|U.d| < |U| \quad \text{or} \quad U.d = U.$$

Letterwise Schreier simulation. For every such U, d,

$$|U.d| = |U| \Longrightarrow U.d = U \quad \text{or} \quad U.d = U.\pi \text{ for some } \pi \in \Pi.$$

Shortest core normal form. Some shortest escape from every reachable unit state consists of a Π-word followed by one defect letter.

The first is local and forbids nontrivial defect transport. The second locally rewrites every such transport. The third is only an existential property of one optimal path.

The equality in letterwise simulation is only an equality of image subsets. The original defect prefix need not remain in the marked-kernel stratum; the normalization theorem constructs an equal-image prefix that does.

Proposition 4.1 (Strict Normalization Hierarchy)**.**

The implications

$$\textit{clean exit} \Longrightarrow \textit{letterwise Schreier simulation} \Longrightarrow \textit{shortest core normal form}$$

are strict.

Proof. The first implication is immediate. For the second, replace every rank-preserving defect step in a shortest escape by the equal zero- or one-letter permutation move; the terminal defect exit remains unchanged and the length does not increase.

For strictness, use $Q = \{0, 1, 2\}$. The binary automaton

$$p = (0, 2, 1), \qquad d = (0, 0, 1)$$

is letterwise simulable but not clean: on $\{0, 2\}$, the letter d gives $\{0, 1\} = \{0, 2\}.p$. For the second strictness, take the identity permutation and

$$d_1 = (0, 0, 1), \qquad d_2 = (0, 2, 0).$$

On $\{0, 1\}$, d_2 transports injectively to $\{0, 2\}$ and cannot be simulated by the identity, while d_1 is an immediate shortest exit. The symmetric statement holds on $\{0, 2\}$. Thus shortest core normal form holds on every reachable nonsingleton although letterwise simulation fails.

4.3 Schreier cover distance

Theorem 4.2 (Schreier Normalization)**.**

Assume letterwise Schreier simulation. For every reachable nonsingleton state T,

$$\omega(T) = 1 + \mathrm{dist}_{\mathcal{G}_K^\Pi}(T, \partial_1 \mathcal{G}_K^\Pi).$$

Proof. Normalize a shortest escape letter by letter. A rank-preserving defect step is replaced by the equal identity move or by one permutation generator. This leaves every intermediate subset unchanged and does not increase length. The normalized prefix lies in $\mathcal{G}_K^\Pi$, and its final letter exits from the cover boundary. This gives a core escape no longer than $\omega(T)$.

Conversely, a shortest permutation path from T to the cover boundary, followed by a defect letter witnessing that boundary, is an escape word. The two inequalities give equality.

Let $G = \langle \Pi \rangle$. Since Π is inverse-closed, its Cayley graph is undirected. The orbit map

$$G \longrightarrow \mathrm{Orb}_G(T), \qquad g \longmapsto T.g,$$

is a surjective labelled graph morphism. Hence

Corollary 4.3 (Cayley-to-Schreier Radius)**.**

Whenever the cover boundary is nonempty,

$$\Delta(T) \leq \operatorname{diam} \mathrm{Sch}(G/\operatorname{Stab}_G(T), \Pi) \leq \operatorname{diam} \mathrm{Cay}(G, \Pi),$$

where $\Delta(T)$ is the maximum distance from the orbit to its cover boundary.

5 Commuting Involution Cores

Assume now that the distinct nonidentity permutation generators are pairwise commuting involutions. Then

$$G \cong (\mathbf{F}_2)^s$$

for some $s \geq 0$.

5.1 Global and statewise Schreier rank

A basis of G can be selected from the declared generators. Every group element is therefore represented by a word of length at most s, giving

$$\operatorname{diam}\operatorname{Cay}(G,\Pi) \leq s.$$

For a subset T, define the quotient rank

$$s_T = \dim_{\mathbf{F}_2} G/\operatorname{Stab}_G(T).$$

The generator images span this quotient, and a basis can again be selected from them. Therefore

$$\Delta(T) \leq s_T \leq s.$$

Orbit-stabilizer gives

$$2^{s_T} = |\operatorname{Orb}_G(T)|.$$

If $|T| = r$, this orbit lies among the r-subsets of Q, so

$$s_T \leq \left\lfloor \log_2 \binom{n}{r} \right\rfloor.$$

Put

$$\sigma_r = \max_{T \in \mathcal{R}_r} s_T.$$

Theorem 5.1 (Statewise Commuting-Core Bound)**.**
For a synchronizing automaton in the declared commuting-involutive letterwise-simulation class,

$$D_{\text{sync}} \leq 1 + \sum_{r=2}^{n-1} (1 + \sigma_r)$$

and

$$D_{\text{sync}} \leq 1 + \sum_{r=2}^{n-1} \left(1 + \min\left\{s, \left\lfloor \log_2 \binom{n}{r} \right\rfloor\right\}\right).$$

Proof. A defect-one letter drops every subset rank by at most one. Thus a reset path visits ranks one at a time. At full rank any defect letter gives an immediate unit drop. At a reachable rank-r state, Theorem 4.2 and the quotient-rank diameter bound give an escape of length at most $1 + s_T$, hence at most $1 + \sigma_r$. Summing these independently valid rank costs proves the first inequality. The orbit-size estimate proves the second.

The uniform consequence is immediate.

Corollary 5.2 (Global Rank Bound)**.**
Under the same hypotheses,

$$D_{\text{sync}} \leq 1 + (n-2)(s+1).$$

5.2 Removing the group parameter

The action $G \leq S_n$ is faithful. Decompose Q into its nontrivial G-orbits $O_1, \dots, O_m$, with

$$|O_i| = 2^{t_i}.$$

Fixed-point orbits contribute no quotient rank and may be omitted. Their action kernels are all of G, so omitting them does not change the kernel of the diagonal action map.

Let K_i be the kernel of the action on O_i. The faithful transitive quotient G/K_i is abelian and therefore regular, so

$$\operatorname{codim}_{\mathbf{F}_2} K_i = t_i.$$

The diagonal map

$$G \longrightarrow \bigoplus_i G/K_i$$

is injective. Hence $s \leq \sum_i t_i$. Since $2^t \geq 2t$ for every $t \geq 1$,

$$n \geq \sum_i 2^{t_i} \geq 2 \sum_i t_i \geq 2s.$$

Lemma 5.3 (Faithful Elementary-Abelian Rank)**.**
Every faithful elementary abelian 2*-subgroup of* S_n *has*

$$s \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

Combining Lemma 5.3 with Corollary 5.2 gives

$$D_{\text{sync}} \leq 1 + (n-2)\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right).$$

For $n = 2m$, the right side is $2m^2 - 1$ and its difference from $(2m-1)^2$ is $2(m-1)^2$. For $n = 2m+1$, it is $2m^2 + m$ and its difference from $4m^2$ is $m(2m-1)$. Both are nonnegative.

Theorem 5.4 (Parameter-Free Černý Corollary)**.**
Every synchronizing automaton in the commuting-involutive, defect-one, letterwise-simulation class satisfies

$$\boxed{D_{\text{sync}} \leq 1 + (n-2)\left(\left\lfloor \frac{n}{2} \right\rfloor + 1\right) \leq (n-1)^2.}$$

This is a quadratic subclass theorem. It is not a universal bound on all synchronizing automata.

6 Binary Wait Number and Clean Hole Dynamics

Assume the alphabet is binary, with one permutation letter p and one defect-one letter d. Inverse closure forces $p^{-1} = p$, so $p^2 = 1$.

Definition 6.1 (Binary Clean Automaton)**.**

A binary clean automaton is a synchronizing binary member of the inverse-closed defect-one structural class of Section 4.1 that satisfies clean exit. Explicitly, $\Sigma = \{p, d\}$, the letter p is a permutation with $p^2 = 1$, $\operatorname{rank}(d) = n-1$, and every reachable nonsingleton subset U satisfies

$$|U.d| < |U| \quad \text{or} \quad U.d = U.$$

6.1 Wait-number normal form

Under letterwise simulation, normalize a shortest reset word by deleting rank-preserving defect loops and replacing every remaining rank-preserving defect step by the equal one-letter p step. Minimality forbids a leading p, a trailing p, and an adjacent pp. Every strict d step reduces the rank by exactly one, so exactly $n-1$ such steps occur.

Theorem 6.2 (Binary Wait-Number Normal Form)**.**

Every shortest normalized reset word has the form

$$d\, p^{\eta_{n-1}} d\, p^{\eta_{n-2}} d \cdots p^{\eta_2} d, \qquad \eta_r \in \{0, 1\}.$$

If

$$\nu(A) = \min \sum_{r=2}^{n-1} \eta_r$$

over shortest normalized reset words, then

$$\boxed{D_{\text{sync}} = n - 1 + \nu(A).}$$

The displayed sum is in fact fixed by the shortest length, but the minimum makes the definition independent of witness selection.

6.2 Defect-one functional graph

A rank-$(n-1)$ map has one missing image state, one double fiber, and all other nonempty fibers singletons. Its functional graph therefore consists of a unique directed tail

$$X = \{x_0 = m, x_1, \dots, x_{h-1} = a\}$$

entering a directed cycle at the double-fiber target, together with possibly other disjoint cycles. If b is the cycle predecessor of the entry point, then the collision pair is

$$C = \{a, b\}.$$

Every state outside X is a cycle vertex. Write

$$c = |Q \setminus X|.$$

For a current subset $T = Q \setminus H$, the letter d drops rank exactly when

$$H \cap C = \emptyset.$$

In that case the next hole set is

$$H^+ = \{m\} \cup d(H).$$

Starting from $H_0 = \emptyset$, induction gives

$$H_q = \{m, d(m), \dots, d^{q-1}(m)\}, \qquad 0 \le q \le h,$$

where the displayed set is empty for $q = 0$. For $q < h$, the tail endpoint $a = d^{h-1}(m)$ has not yet entered H_q, while b is a cycle vertex. Therefore

$$H_q \cap C = \emptyset \quad (q < h), \qquad H_h = X, \qquad X \cap C = \{a\}.$$

Thus the first h applications of d are strict drops and the next application is the first plateau. If a wait p precedes the drop, this becomes

$$H^+ = \{m\} \cup d(p(H)), \qquad p(H) \cap C = \emptyset.$$

6.3 The one-wait obstruction

Assume now the stronger clean-exit condition.

Starting from Q, repeated d removes the tail in order. After h strict drops, the hole set is X and the surviving subset $Q \setminus X$ is the union of all cycle vertices. If $c = 1$, this is already a singleton.

Suppose $c > 1$. The cycle union is fixed by d. Synchronization forces its p-image to lie on the d-drop boundary. Otherwise clean exit fixes both subsets under d, while p toggles them, giving a closed nonsingleton trap. Thus

$$p(X) \cap C = \emptyset,$$

and one wait followed by d gives a strict drop.

After this wait, let H_q be the hole set after the q letters in the canonical suffix d^q. As long as those letters are strict drops, induction on the hole recurrence gives

$$H_q = \{m, d(m), \dots, d^{q-1}(m)\} \cup d^q(p(X)).$$

Theorem 6.3 (Binary Clean One-Wait Theorem)**.**

If $c > 1$, every letter in the canonical suffix d^{c-1} after $d^h p$ is a strict drop until a singleton is reached. Consequently $d^h p d^{c-1}$ is a reset word and $\nu(A) \le 1$.

Proof. Suppose that after $q \ge 1$ strict drops in the canonical suffix, and before reaching a singleton, the next d is rank preserving. Put $H' = H_q$ and $T' = Q \setminus H'$. Clean exit gives $T'.d = T'$. An invariant subset of a finite functional graph is a union of cycles. Therefore

$$H' = X \cup Y$$

for a union Y of complete d-cycles.

The tail endpoint a belongs to X, so $p(a) \in p(X)$. The recurrence puts $d^q(p(a))$ in H'. If $p(a)$ is a tail state, it already belongs to X. If it is a cycle state, then $d^q(p(a))$ is on the same cycle; because Y is a union of complete cycles, $p(a) \in Y$. In both cases $p(a) \in H'$, and therefore

$$a \in p(H') \cap C.$$

Thus d cannot drop rank from $p(T')$. Clean exit fixes $p(T')$ under d as well. The letters d and p now preserve a closed set consisting of one or two nonsingleton subset-states, namely $\{T', p(T')\}$. This contradicts synchronization.

Corollary 6.4 (Exact Binary Clean Depth)**.**
With $c = |Q \setminus X|$,

$$\nu(A) = \begin{cases} 0, & c = 1, \\ 1, & c > 1, \end{cases} \qquad D_{\text{sync}} = \begin{cases} n-1, & c = 1, \\ n, & c > 1. \end{cases}$$

Shortest constructive words are d^h when $c = 1$ and $d^h p d^{c-1}$ when $c > 1$.

Every reset word needs $n-1$ defect-one drops. If $c > 1$, d alone is trapped on a nonsingleton cycle union, so at least one p is also necessary. This proves the lower bounds in Corollary 6.4.

An explicit family realizes the second case for every $n \geq 3$. On $Q_n = \{0, 1, \dots, n-1\}$, let p_n swap $n-2$ and $n-1$ and fix all other states, and define

$$d_n(0) = 0, \qquad d_n(i) = i+1 \quad (1 \leq i \leq n-3), \qquad d_n(n-2) = 0, \qquad d_n(n-1) = n-1.$$

The intermediate range is empty when $n = 3$. The reachable nonsingletons are

$$T_0 = Q_n, \qquad T_k = Q_n \setminus \{1, \dots, k\} \quad (1 \leq k \leq n-2), \qquad U = \{0, n-2\}.$$

For $k < n-2$, both transposed states belong to T_k, so p_n fixes T_k and d_n drops it to T_{k+1}. The set $T_{n-2} = \{0, n-1\}$ is fixed by d_n, p_n sends it to U, and d_n sends U to the singleton $\{0\}$. This also proves clean exit on the complete reachable orbit. The word $d_n^{n-2} p_n d_n$ has length n, and Corollary 6.4 proves that it is shortest. Therefore:

Corollary 6.5 (Exact Clean-Class Extremal Depth)**.**
Let $D^{\text{clean}}_{\max}(n)$ be the maximum reset depth over synchronizing binary clean automata on n states. Then

$$D^{\text{clean}}_{\max}(2) = 1, \qquad D^{\text{clean}}_{\max}(n) = n \quad (n \geq 3).$$

7 Exact Finite Certificates

The computations use finite sets, integer maps, exact subset breadth-first search, and exact partition comparisons. No floating-point tolerance enters the certificates.

7.1 Structural census

The complete supplied synchronizing binary censuses at $n = 3, 4$ give:

n	structural base	clean	letterwise simulation	shortest normal form
3	12	8	12	12
4	44	8	12	12

The equality of the simulation and normal-form counts is a finite coincidence; Proposition 4.1 proves that the two notions are distinct.

7.2 Complete binary clean exhaustion

Complete labelled exhaustion of every involution/defect-one pair through $n = 6$ gives:

n	clean synchronizing pairs	$\nu = 0$	$\nu = 1$	maximum depth
2	4	4	0	1
3	24	12	12	3
4	96	48	48	4
5	600	240	360	5
6	3600	1440	2160	6

These data independently replay Theorem 6.3 and Corollary 6.5 in the declared finite range. They are not the proof of either statement.

7.3 Multi-generator fixture

The exact positive fixture

$$p_1 = (1, 0, 3, 2, 4),$$
$$p_2 = (2, 3, 0, 1, 4),$$
$$d = (0, 1, 2, 4, 4)$$

has $n = 5$, elementary-abelian core rank $s = 2$, and group order 4. Its reset depth is 7. The global bound in Corollary 5.2 is 10, while the statewise quotient-rank profile gives 9. It is a nondegenerate instance with $s = 2$.

8 Interpretation and Open Boundary

The result separates several objects that should not be conflated:

$$\begin{aligned}
\text{pair distance} &\neq \text{marked-kernel state},\\
\text{clean exit} &\neq \text{letterwise simulation},\\
\text{letterwise simulation} &\neq \text{existential normal form},\\
\text{global group rank } s &\neq \text{subset quotient rank } s_T,\\
\text{rankwise accounting} &\neq \text{path-compatible potential},\\
\text{finite certificate} &\neq \text{theorem proof}.
\end{aligned}$$

The binary clean one-wait proof uses the fact that rank-preserving defect steps are literal loops. It does not extend automatically to letterwise simulation, where a defect letter may transport the current subset to a different root inside the same Schreier orbit. The most immediate open problem is to determine whether an analogous bounded-wait theorem holds for a natural strict superclass of clean binary automata.

For several noncommuting permutation generators, the exact theorem remains

$$\Delta(T) \le \operatorname{diam} \operatorname{Sch}(G/\operatorname{Stab}_G(T), \Pi),$$

but a useful uniform bound requires additional group structure. Another open direction is to combine the high-capacity fiber-incidence branch with the Schreier-controlled unit branch in automata whose reachable rank layers contain both kinds of states.

No claim is made about arbitrary synchronizing automata, the standard Černý family, random automata, probabilistic synchronization, quantum channels, or operator mortality.

9 Claim Status and Boundary

Surface	Evidence level	Support
Pair-hitting identity	Theorem	manuscript proof; Lean 4 formalization; exact replay
Packing/common-transversal equivalence	Theorem	manuscript proof; exact replay
Waiting-capacity interface	Theorem	manuscript proof; exact rational replay
Marked-kernel corridor and kernel cover	Theorem	manuscript proof; exact replay
Reachable-unit packing inversion and area identity	Theorem	manuscript proof; exact replay
Strict normalization hierarchy	Theorem	explicit finite witnesses
Schreier normalization and Cayley quotient bound	Theorem	manuscript proof; exact replay
Statewise and global commuting-core bounds	Theorem	manuscript proof; exact replay
$s \le \lfloor n/2 \rfloor$ and parameter-free Černý corollary	Theorem	manuscript proof; conditional arithmetic step formalized in Lean 4; exact replay
Binary wait-number normal form	Theorem	manuscript proof; exact replay
Binary clean one-wait theorem and exact extremal depth	Theorem	manuscript proof; exact replay
$n = 3, 4$ corridor census and $n \le 6$ clean exhaustion	Computational Certificate	exact finite replay
bounded-wait extension beyond clean exit	Research Program	open

The Lean package in Appendix A fully formalizes Theorem 2.1 and formalizes the arithmetic implication used in Theorem 5.4 conditional on its group-rank and reset-depth inputs. It does not formalize the group-action bound $s \le \lfloor n/2 \rfloor$, the automata-theoretic commuting-core bound, or any other manuscript theorem.

10 Conclusion

Rank escape in a synchronizing automaton is exactly pair hitting, and a long unit wait is exactly a corridor confined to one marked transformation kernel. When a permutation core controls that corridor, the relevant geometry is a subset Schreier graph rather than the unrestricted rank layer.

In the Schreier-controlled classes studied here, rank-preserving transport is normalized by a declared permutation action. Outside that setting, marked-kernel persistence alone need not control how useful future kernel coarsenings become; such path-compatible transport questions lie beyond this paper.

For commuting involution cores, elementary-abelian quotient rank gives both statewise and global reset bounds. Faithfulness removes the group parameter and places the entire declared class below the Černý threshold. In the binary clean specialization, hole dynamics is stronger: a second plateau on the canonical descent would create a closed nonsynchronizing trap. The reset depth is therefore exactly $n-1$ or n, with extremal value n for every $n \geq 3$.

Appendix A Computational Artifacts

Computational sources and exact records are available in the RIME repository under `experiments/paper23/`; paths below are relative to that directory.

Role	Repository path
pair hitting and packing	`pair_hitting.py`
fiber-incidence layer	`fiber_incidence_potential.py`; `fiber_incidence_controls.py`
marked-kernel corridors	`kernel_schreier_corridors.py`
complete clean exhaustion	`enumerate_binary_clean_corridors.py`
finite census records	`results/n2_k2_v2.json`; `results/n3_k2_v2.json`
merged four-state census	`results/n4_k2_v2_summary.json`; `results/n4_k2_v2_shards/`

The listed computations replay the finite certificates; they do not replace the proofs in Sections 2–6.

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